

Database Fundamentals Quiz Questions

1. What is data described as in the lecture?

- A. Processed and organized facts
- B. Raw, unorganized facts that need processing
- C. Structured information ready for use
- D. Meaningful context without organization

Answer: B

2. When data is processed and organized to make it useful, what is it called?

- A. Raw facts
- B. Database
- C. Information
- D. Metadata

Answer: C

3. Which example from the lecture illustrates data turning into information?

- A. Website traffic by country analyzed for trends
- B. Random temperature readings without analysis
- C. Unorganized visitor numbers
- D. Decreasing traffic from the U.S.

Answer: A

4. According to the lecture, what is a database?

- A. A collection of raw facts
- B. A collection of information organized for easy access, management, and update
- C. A spreadsheet for storing data
- D. A text file with unprocessed data

Answer: B

5. How is data in a database typically organized?

- A. Randomly without structure
- B. To model relevant aspects of reality supporting processes
- C. Only in text files
- D. Without any updates allowed

Answer: B

6. What alternatives to databases are mentioned in the "Why Database?" section?

- A. Storing data in a text file or spreadsheet
- B. Using cloud storage only
- C. Manual paper records
- D. Email attachments

Answer: A

7. Which is an advantage of using a database?

- A. Increasing redundancy
- B. Reducing redundancy
- C. Making data incompatible
- D. Avoiding data sharing

Answer: B

8. How can databases help avoid inconsistency?

- A. By allowing multiple versions of data
- B. Through organized structure and management
- C. By storing data in separate files
- D. Ignoring integrity constraints

Answer: B

9. What advantage allows multiple users to access the same data?

- A. Data can be shared
- B. Security restrictions are removed
- C. Redundancy is increased
- D. Backup is unnecessary

Answer: A

10. Which advantage involves enforcing rules like data validity?

- A. Providing backup and recovery
- B. Enforcing integrity constraints
- C. Reducing redundancy
- D. Applying security restrictions

Answer: B

11. What is a disadvantage of databases mentioned in the lecture?

- A. They are inexpensive to use
- B. They need expertise which is expensive
- C. They always compatible with other systems
- D. They reduce costs overall

Answer: B

12. Why is DBMS considered expensive?

- A. The software itself is costly
- B. It requires no maintenance
- C. It is free for all users
- D. It simplifies all processes

Answer: A

13. What compatibility issue is noted as a disadvantage?

- A. DBMS may be incompatible with other DBMS
- B. All DBMS work seamlessly together
- C. Databases share data easily across systems
- D. No expertise is needed for integration

Answer: A

14. What composes a database system according to the lecture?

- A. Only the database
- B. The database and the software (DBMS and application programs)
- C. Just application programs
- D. Metadata alone

Answer: B

15. Which type of database is listed first in the types of databases?

- A. Distributed database
- B. Centralized database
- C. NoSQL database
- D. Cloud database

Answer: B

16. Which database type is mentioned as supporting unstructured data?

- A. Relational database
- B. NoSQL database
- C. Personal database
- D. Operational database

Answer: B

17. What is an example of a continued type of database in the lecture?

- A. Graph database
- B. End-user database

- C. Commercial database
- D. Centralized database

Answer: A

18. Where can more info on database types be found as per the lecture?

- A. <https://www.tutorialspoint.com/Types-of-databases>
- B. A local library
- C. The DBMS manual
- D. SQL Server documentation

Answer: A

19. What is a DBMS?

- A. System software for creating and managing databases
- B. A type of database
- C. Raw data collection
- D. Application program only

Answer: A

20. How does DBMS serve users and programs?

- A. As an interface ensuring organized and accessible data
- B. By deleting data automatically
- C. Ignoring updates
- D. Limiting access to one user

Answer: A

21. Which DBMS function involves creating the structure?

- A. Manipulating database
- B. Defining database
- C. Backup & recovery
- D. Performance

Answer: B

22. What function handles inserting, updating, and deleting data?

- A. Constructing database
- B. Manipulating database
- C. Data security & integrity
- D. Concurrency

Answer: B

23. Which function ensures multiple users can access without issues?

- A. Backup & recovery
- B. Concurrency
- C. Data dictionary
- D. Defining database

Answer: B

24. What is metadata in the context of database systems?

- A. Raw user data
- B. Stored DB definition (data dictionary)
- C. Application programs
- D. Queries only

Answer: B

25. In the database system diagram, what do users send to application programs?

- A. Data directly
- B. Queries
- C. Metadata
- D. Software updates

Answer: B

26. What layer in database architecture includes business rules like salary limits?

- A. Data layer
- B. Business layer
- C. Presentation layer
- D. Metadata layer

Answer: B

27. In the Intelligent Client (2-Tier) architecture, what is on the client side?

- A. Data only
- B. Presentation and business
- C. Data and business
- D. Presentation only

Answer: B

28. What does the N-Tier architecture separate?

- A. Presentation, business, and data
- B. Only data and software
- C. Users and queries
- D. Metadata and integrity

Answer: A

29. What provides a logical representation of database structure?

- A. Physical data model
- B. Logical data model
- C. Presentation model
- D. Business model

Answer: B

30. What describes how data is stored and accessed in the computer?

- A. Logical data model
- B. Physical data model
- C. Metadata model
- D. Query model

Answer: B

31. What does SQL stand for?

- A. Structured Query Language
- B. System Query Logic
- C. Simple Query Language
- D. Standard Query Layer

Answer: A

32. What standard is SQL based on?

- A. ISO
- B. ANSI
- C. IEEE
- D. W3C

Answer: B

33. Why use SQL according to the lecture?

- A. To access data in relational DBMS
- B. To delete all databases
- C. Only for creating views
- D. Ignoring permissions

Answer: A

34. What can SQL do with records in a database?

- A. Insert, select, update, delete
- B. Only select
- C. Only insert
- D. Delete and create only

Answer: A

35. What is Microsoft SQL Server?

- A. A relational database management system
- B. A text editor
- C. A spreadsheet tool
- D. A web browser

Answer: A

36. What is the core component of Microsoft SQL Server?

- A. SQL Server Database Engine
- B. Application programs
- C. Metadata dictionary
- D. Presentation layer

Answer: A

37. What does the SQL Server Database Engine manage?

- A. Only queries
- B. Data storage, processing, and security
- C. Business rules
- D. User interfaces

Answer: B

38. What are created and executed by the Database Engine?

- A. Stored procedures, triggers, views
- B. Text files

- C. Spreadsheets
- D. Emails

Answer: A

39. What is SQL Server Management Studio (SSMS)?

- A. An integrated environment for managing SQL infrastructure
- B. A database type
- C. A query language
- D. A data model

Answer: A

40. What can SSMS be used for?

- A. Access, configure, manage, and develop SQL Server components
- B. Only install SQL Server
- C. View images only
- D. Process text files

Answer: A

Wish you all the best!

Fathy.